



STRATHMORE SCHOOL OF COMPUTING AND ENGINEERING SCIENCES

BACHELOR OF SCIENCE CYBERSECURITY, NETWORK AND SECURITY
BACHELOR OF SCIENCE INFORMATION AND COMPUTER SCIENCE

CONTINUOUS ASSESSMENT TEST (CAT TWO)

CNS 1205&ICS 1205: LINEAR ALGEBRA

DATE:30/09/2024

DUE 18/12/2024-12.00 P.M

INSTRUCTIONS

1. ANSWER ALL QUESTION

QUESTION ONE (30Marks)

1. (a) Write a simple algorithm in R-studio that can be used to solve the system of linear equations. (5marks)

$$x_2 + x_3 - 2x_4 = -3$$

$$x_1 + 2x_2 - x_3 = 2$$

$$2x_1 + 4x_2 + x_3 - 3x_4 = -2$$

$$x_1 - 4x_2 - 7x_3 - x_4 = -19$$

- (b) Find the inverse of the matrix A using Gaussian -Jordan elimination method. (5 marks)

$$\begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ -6 & 2 & 3 \end{pmatrix}$$

- (c) Find the eigenvalues and corresponding eigenvectors of the matrix . (6marks)

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -4 \end{bmatrix}$$

- (d) i Write the uncoded row matrices of size (1×3) for the message **MEET ON SUNDAY** . (3 marks)
ii Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message **MEET ON SUNDAY**. (6 marks)

- (e) Use Cramer's Rule to solve the system of linear equations for x, y, z . (5 marks)

$$-x + 2y - 3z = 1$$

$$2x + z = 0$$

$$3x - 4y + 4z = 2$$

- (f) Find an equation of the line passing through the points (2,4) and (-1,3).Using determinant techniques (4 marks)
- (g) Find the area of the triangle whose vertices are (-1,2),(2,2) and (2,4) (5 marks)
- (h) A country is divided into 3 regions. Each year , 10% of the residents of region 1 move to region 2 and 5% move to region 3; 15% of the residents of region 2 move to region 1 and 5% move to region 3 ; and 10% of the residents of region 3 move to region 1 and 10% move to region 2. This year each region has a population of 100000. Find the population of each region (a) 1 year (b) in 3 years (6 marks)
- (i) Find the least squares regression line for the points (2,1),(3,3),(4,2),(5,4) and (6,4) (5 marks)